

T1 Motion, Forces and Energy (1.1 - 1.4)

1.1 Physical quantities and measurement techniques

Measuring length:

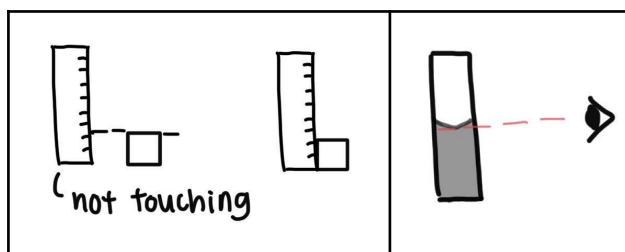
Instrument	Range of length	Precision
measuring tape	$l > 100\text{cm}$	0.1cm
metre rule	$5\text{cm} < l < 100\text{cm}$	0.1cm
vernier callipers	$1\text{cm} < l < 10\text{cm}$	0.01cm
micrometer screw gauge	$l < 2\text{ cm}$	0.001cm

Zero error:

- When equipment isn't reset to 0 properly before using it
 - To minimise, reset to 0 before use

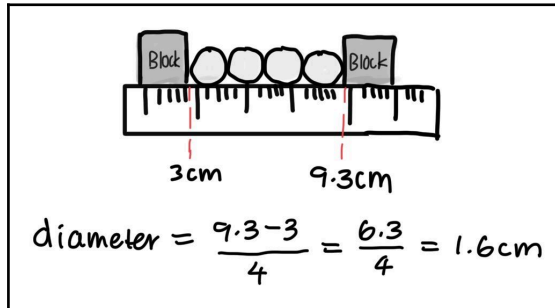
Parallax error:

- Occurs during measurement -> incorrect position of eyes or object not touching marking of scale
 - To minimise, read at perpendicular angle
 - Read at perpendicular angle at bottom of meniscus
 - Ensure object is in contact with the scale



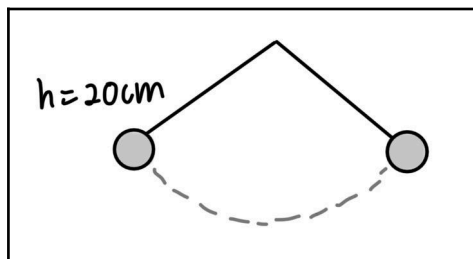
More measurement techniques

Measuring diameter of a ball bearing:



- Blocks mark edges of the first and last ball bearing
- Divide the total length of ball bearings, divided by the total number of ball bearings

Measuring period of a pendulum:



- Drop the bob at a height of 20cm
- Immediately start stopwatch, and record the time taken for 10 complete oscillations
- Calculate period of 1 oscillation: Time taken / 10

Physical quantities:

- Scalar Quantity - quantity that only has magnitude
- Vector Quantity - quantity that has both magnitude and direction

Scalar quantities: - Distance - Speed - Time - Mass - Density - Temperature - Energy	Vector quantities: - Displacement - Velocity - Acceleration - Mass and weight - Momentum - Electric field strength - Gravitational field strength
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1.2 Motion

Distance and displacement:

Distance - the total length of the path traveled by an object

- Scalar quantity
- Unit: metre (m)

Displacement - the directed distance from the start to the end point, travelled by an object

- Vector quantity
- Unit: metre (m)

Speed and velocity

Speed: the distance travelled per unit time

- Scalar quantity
- Unit: metre per second (m/s)

Velocity: speed in a given direction

- Vector quantity
- Unit: metre per second (m/s)

Acceleration

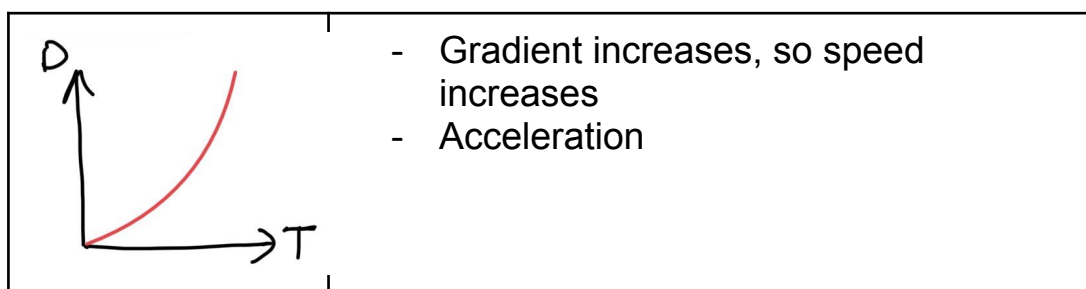
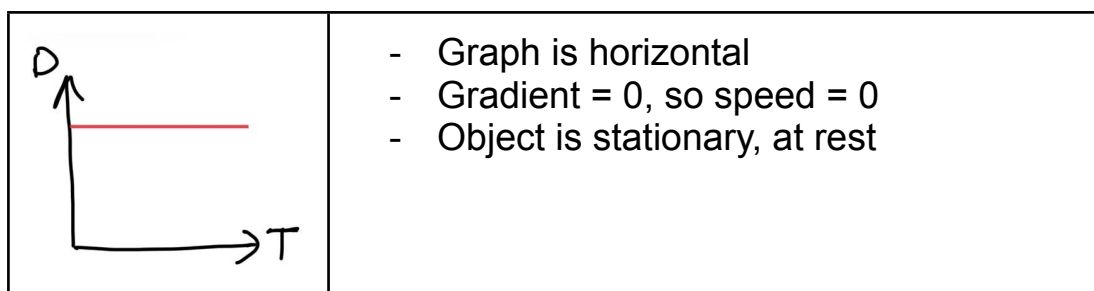
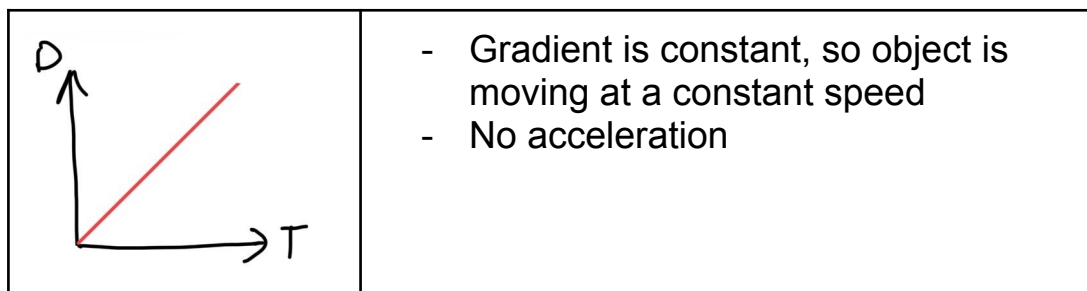
Acceleration: change in velocity per unit time

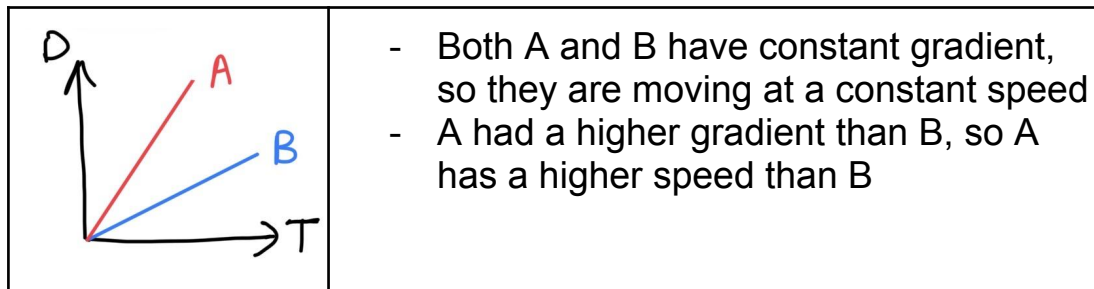
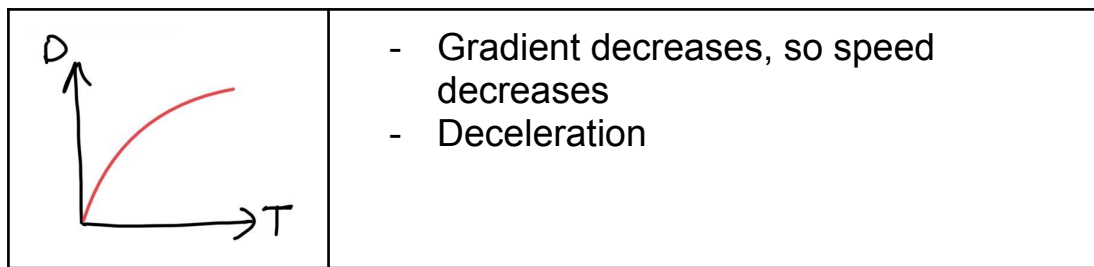
- Vector quantity
- Unit: metre / second² (m/s²)

Deceleration: negative acceleration, change in decreasing velocity per unit time

Distance-time graphs:

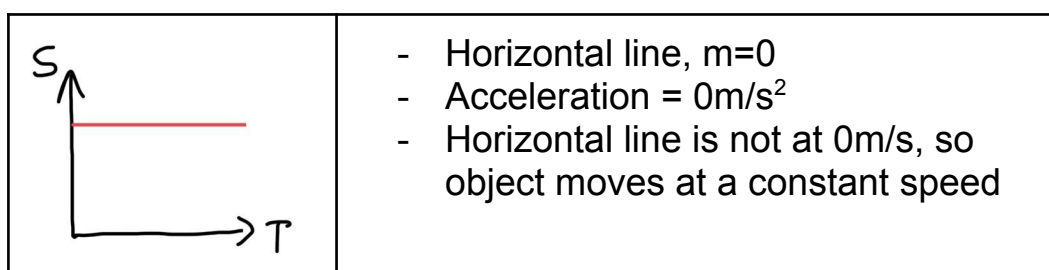
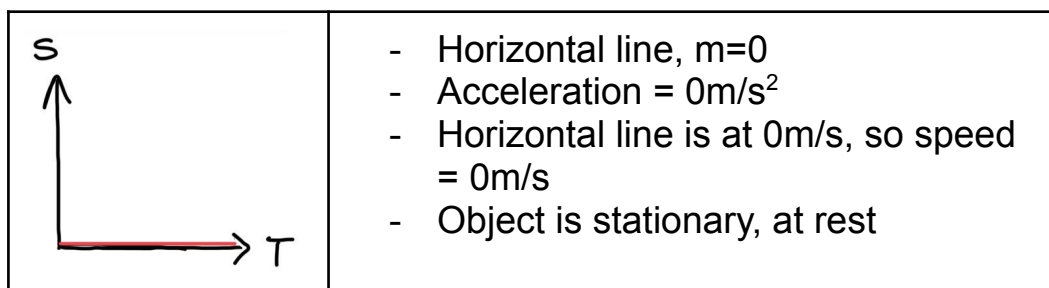
- Gradient = speed

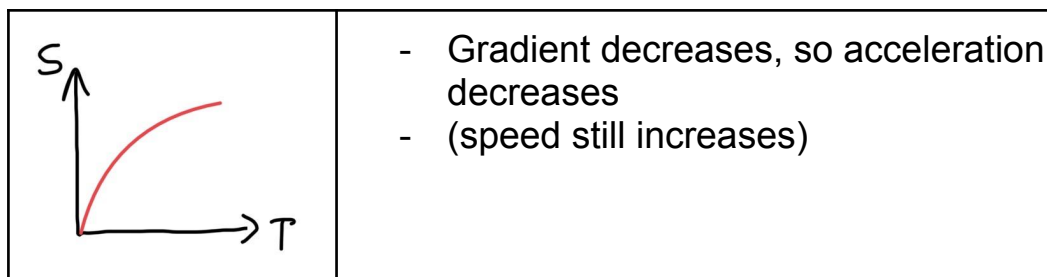
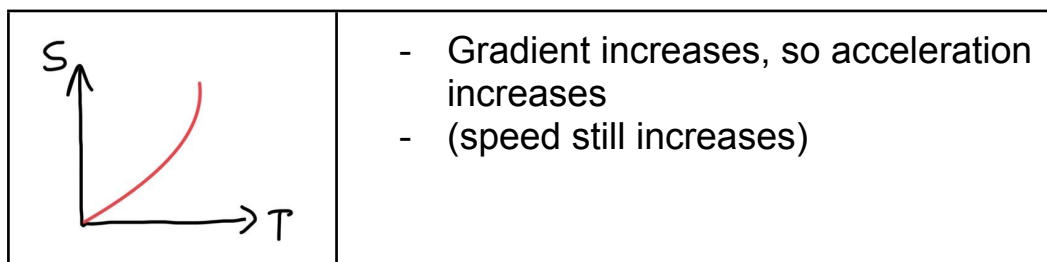
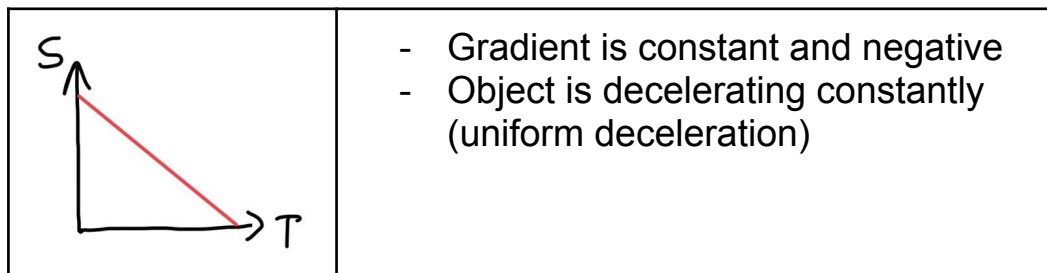
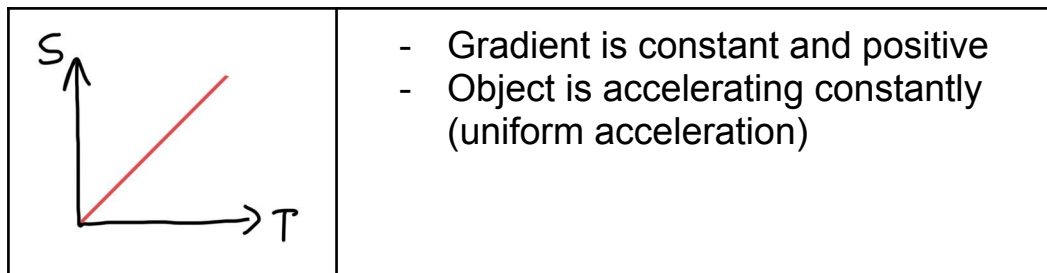




Speed-time graphs:

- Gradient = acceleration





Free fall

- When the motion of an object is under the influence of gravity only
- There is no resistance, and only weight acts on the object
- There is constant acceleration towards the ground due to gravity, which is 9.8m/s^2

NOTE: 9.8m/s^2 acceleration meaning speed increases by 9.8m/s every second.

- This means that every second, distance travelled in that second increases by 9.8m .
- $4.9\text{m} \rightarrow 14.7\text{m} \rightarrow 24.5\text{m} \rightarrow 34.3\text{m}$

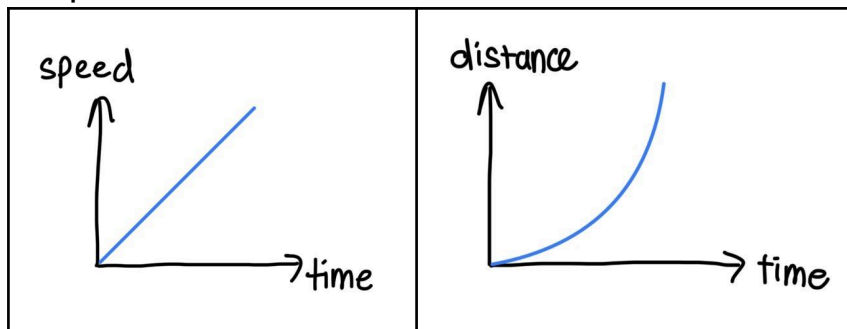
For example - in air:

- When a feather and bowling ball is dropped at the same time at the same height in air, bowling ball reaches the ground first.
- This is because air resistance has a greater effect on the feather, due to its large surface area

In a vacuum:

- When they are both dropped at the same time from the same height in a vacuum, they both have the same speed and reach the ground at the same time
- This is because air resistance has no effect of them in a vacuum - objects are both undergoing free fall

Graphs:

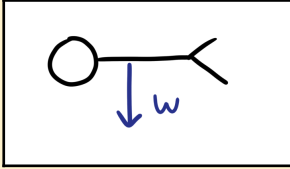


Terminal velocity

- The maximum speed an object reaches when moving through a fluid, due to balanced forces (no resultant force)
- Drag force - liquid/water resistance or air resistance

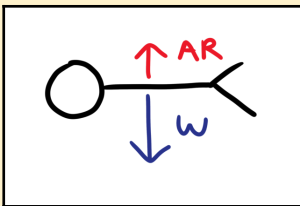
Terminal velocity of a skydiver:

Stage 1:



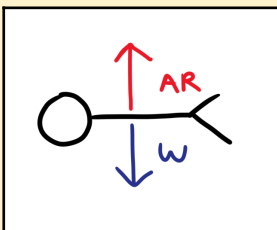
- When skydiver jumps out of the plane, the only force acting on them is weight
- Resultant force = weight, downwards direction
- Skydiver initially accelerates at 9.8m/s^2

Stage 2:



- As the skydiver accelerates, the speed increases, so air resistance also increases
- Resultant force decreases, but still acts downwards
- Skydiver continues to decelerate but at a slower rate

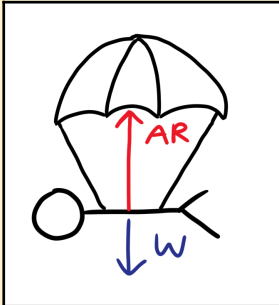
Stage 3:



- Air resistance increases until it is equal to the weight

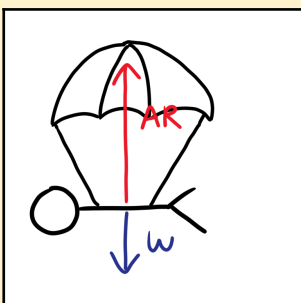
- Forces are balanced, no resultant force
- Acceleration = 0, constant speed.
- Skydiver is facing terminal velocity

Stage 4:



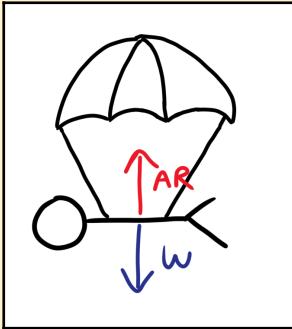
- When parachute opens, large surface area of the parachute collides with many air particles, so air resistance increases
- Air resistance is greater than weight, so resultant force acts upwards, but skydiver continues falling downwards
- Skydiver decelerates

Stage 5:



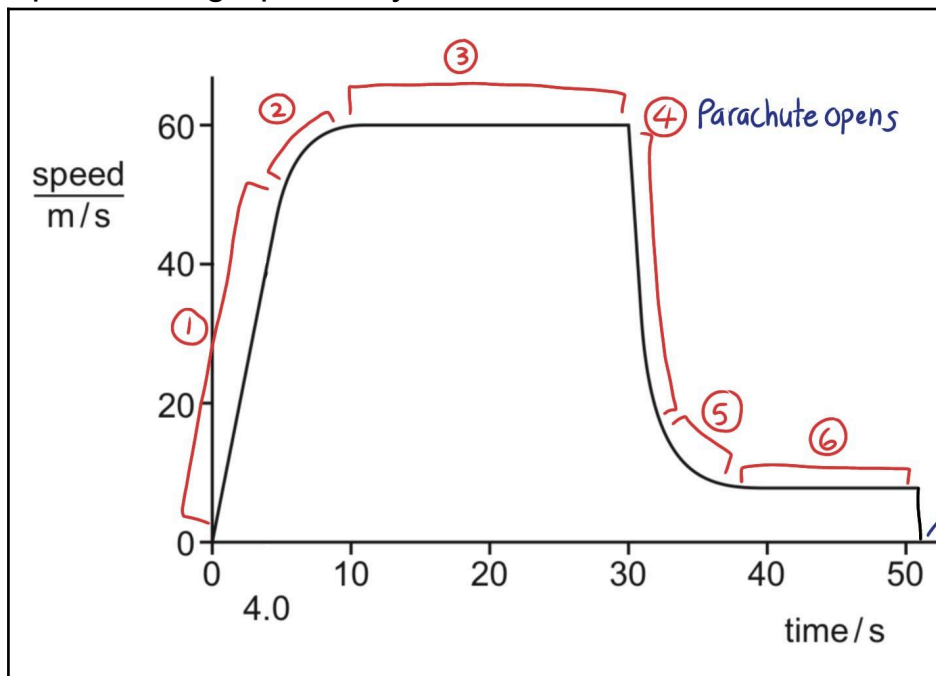
- As the skydiver decelerates, the speed decreases. Air resistance also decreases
- Resultant force decreases, but still acts upwards
- Skydiver continues to decelerate, but at a slower rate

Stage 6:



- Air resistance decreases until it is equal to the weight
- Forces are balanced, no resultant force
- Acceleration = 0, constant speed
- Skydiver is facing terminal velocity

Speed time graph of skydiver:



Difference between free fall and terminal velocity

Free fall:

- Object is falling through vacuum
- Graph is straight, constant gradient as it is constant acceleration of 9.8m/s^2
- Only weight acts on the object

Terminal velocity:

- Object is falling through the air
 - When air resistance increases until it is equal to weight, so resultant force = 0
 - Graph is horizontal, gradient = 0, acceleration = 0.
 - Object moves at a constant speed
-

1.3 Mass and weight

Mass: a measure of the quantity of matter in an object, resisting changes in its state of rest or motion

- Scalar quantity
- Unit: kilogram (kg)
- Remains constant anywhere
- Measured using electric balance

Weight: the gravitational force on an object that has mass

- Vector quantity
- Unit: Newton (N)
- Varies based on gravitational field strength
- Measured using newton meter

Gravitational field strength: the force per unit mass

$$W = mg$$

weight (N) = mass (kg) × gravitational field strength (N/kg)

↓
9.8 (on earth)

- Gravitational field strength (g) on earth's surface = 9.8N/kg
- g is equal to acceleration of freefall, which is 9.8m/s^2
- g is a vector quantity, which points towards the centre of mass

Concept of weight and mass on different planets:

- Gravitational field strength is different
- So, weight will be different

Earth	moon
	
$g = 9.8 \text{ N/kg}$	$g = 1.6 \text{ N/kg}$
$m = 40 \text{ kg}$	$m = 40 \text{ kg}$
$W = 40 \times 9.8$	$W = 40 \times 1.6$
$= 392 \text{ N}$	$= 64 \text{ N}$

1.4 Density

Density: mass per unit volume

$$\rho = \frac{m}{V}$$

density (kg m³) mass (kg) volume (m³)

- Density is a specific value for each material
- The same material has the same density

Density determines whether object floats or sinks:

- If density of object $>$ density of liquid, object sinks
- If density of object $<$ density of liquid, object floats
- If density of object = density of liquid, object floats, but is fully submerged

Density of water - 1g/cm^3 or 1000kg/m^3

To determine density of a regular object:

- Measure mass of object using electric balance
- Measuring length, width, and height, then calculate the volume: $\text{length} \times \text{width} \times \text{height}$
- Calculate density: $\text{density} = \text{mass} / \text{volume}$

To determine density of an irregular object:

- Measure mass of irregular object using electronic balance
- Add 20cm³ water into a measuring cylinder
- Add the irregular object into the measuring cylinder
- Record the volume on the measuring cylinder as V
- Calculate volume of object: $V - 20$
- Calculate density: $\text{density} = \text{mass} / \text{volume}$

To determine volume and density if irregular object floats:

- Measure mass of irregular object using electronic balance
 - Add 20cm³ water into a measuring cylinder
 - Tie a thread to a stone, and place it into the water
 - Measure this volume. Record as V1
 - Remove thread and stone, and tie the irregular object to the other end of the thread
 - Place the irregular object, stone, and thread into the measuring cylinder.
 - Measure this volume. Record as V2
 - Calculate volume of object: $V2 - V1$
 - Calculate density: $\text{density} = \text{mass} / \text{volume}$
-

1.5 Forces

When forces act on an object, they cause the object to change:

- Its size
- Its shape
- Its velocity:
 - Its direction
 - Its speed

Types of forces

Contact forces - forces that act between 2 objects that are physically touching each other

- Examples: pushing/pulling force, normal reaction force, tension, buoyant force

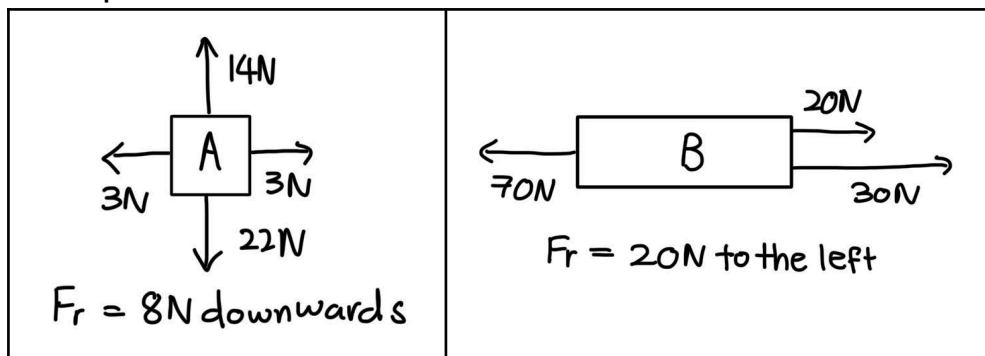
Non-contact forces - forces that act between 2 objects which are not physically touching each other

- They act through a field (invisible region of space), which surrounds an object and exerts a force on other objects in that field
 - Examples: Weight, magnetic force, electrostatic force
-

Resultant force: (F_r)

- The force which has the same effect as all the other forces acting on an object combined
 - If resultant force = 0, balanced forces
 - If resultant force $\neq$ 0, unbalanced forces

Example:



Newton's 1st law of motion:

- When the forces on an object are balanced (no resultant force), object is stationary or moving at a constant velocity, and its acceleration is 0

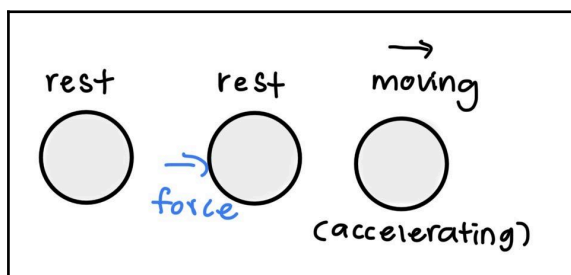
Example -> car moving at constant speed, terminal velocity

Newton's 2nd law of motion

- If there is a resultant force, object will accelerate in the direction of the resultant force

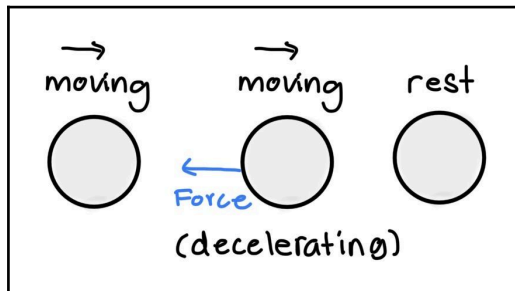
On an object at rest:

- object starts to move, then accelerates, increasing its speed



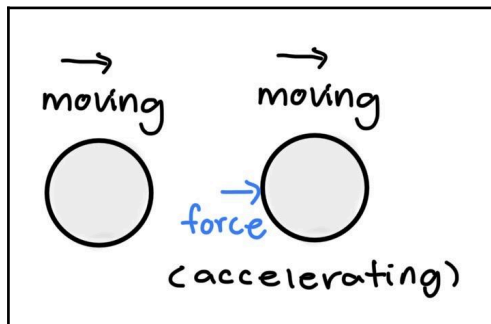
On an object in the opposite direction of its motion:

- object will decelerate until it comes to a stop



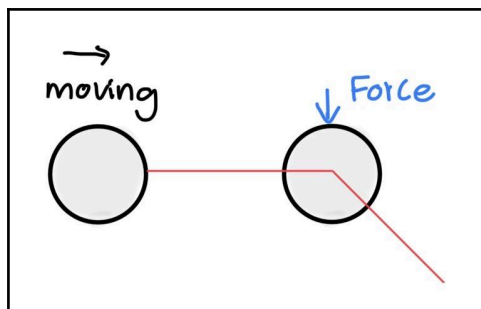
On an object in the same direction of its motion:

- object will accelerate, further increasing its speed



On an object, at a perpendicular angle to its direction of motion:

- object will change direction while speed remains the same



More about acceleration:

- Acceleration of an object is directly proportional to the resultant force
- Acceleration is inversely proportional to mass of the object

$$F = ma$$

force (N) mass (kg) acceleration (m/s²)

Hooke's law

- The extension of an elastic object is directly proportional to the force applied, up to the limit of proportionality

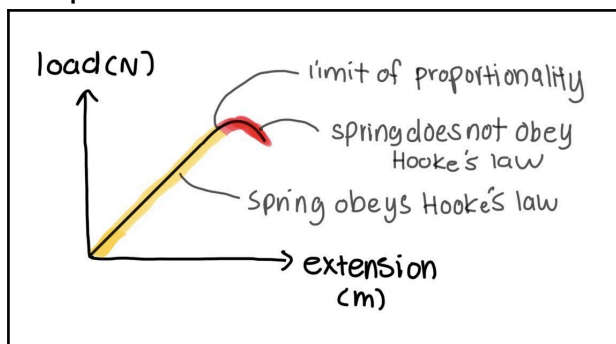
$$F = kx$$

force (N) spring constant (N/m) extension (m)

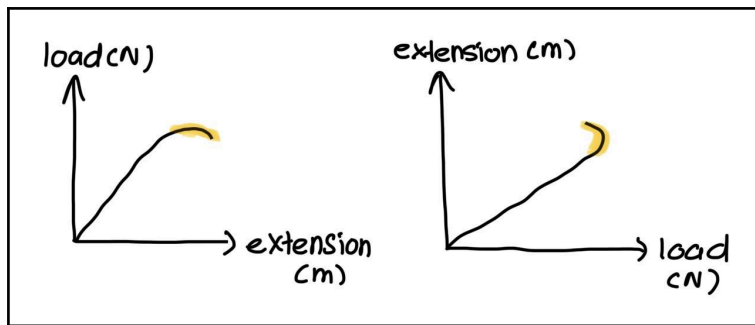
Spring constant: force per unit extension

Limit of proportionality: the final point at which extension is directly proportional to the load

Graph:



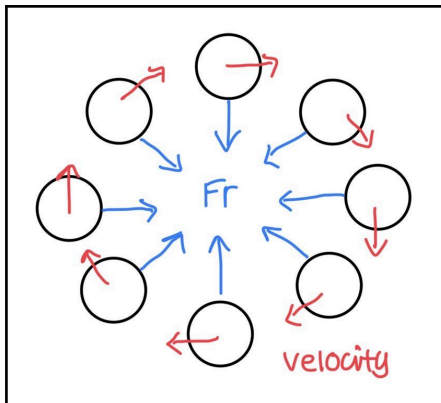
- The graph will always lean towards extension (x) after limit of proportionality:



Investigate how extension varies with applied forces:

- Measure original length of the unstretched spring using a ruler. Record as l_0
- Hang a load of 1N on the spring, then measure the length of the spring. Record as l .
- Calculate extension ($l - l_0$)
- Repeat experiment but with loads of 2N, 3N, 4N and 5N

Circular motion

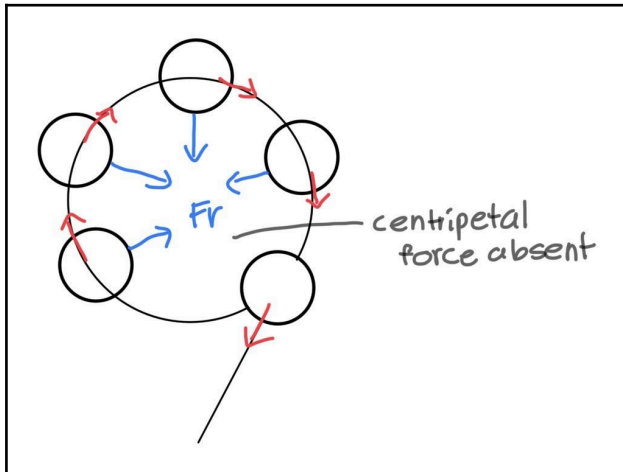


- When a resultant force acts on an object at a perpendicular angle to the direction of its motion, this causes the object to change direction while speed remains constant.
- If resultant force acts on the ball continuously, the continuous change in direction of the object's motion causes the object to move in a circular path.

Centripetal force:

- The resultant force is called the centripetal force
- It acts perpendicular to the direction of motion of the object, and acts towards the centre of the circle
- It causes an acceleration towards the centre of the circle - this is called 'centripetal acceleration'

What happens when the force is absent:



- When centripetal force is absent, object will escape from the circular path along a tangent line, at the point where it leaves the circle
-

Factors affecting resultant force:

Speed: (when mass and radius constant)

- When speed increases, resultant force increases

Radius: (when mass and speed constant)

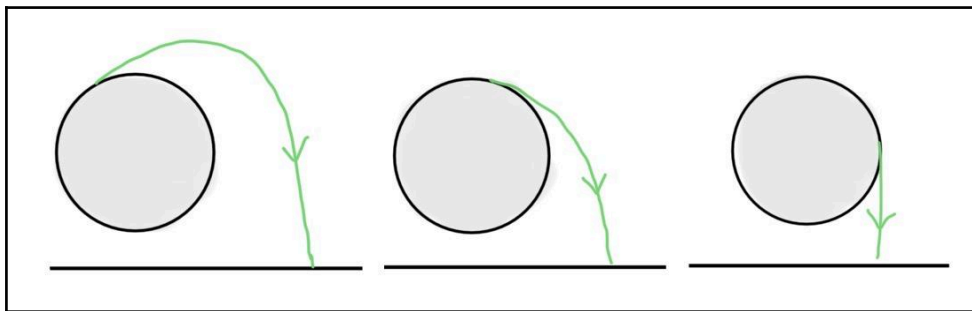
- When radius increases, resultant force decreases

Mass:

- An increased mass needs an increased force to keep speed and radius constant
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Vertical circular motion - ball attached to spring:

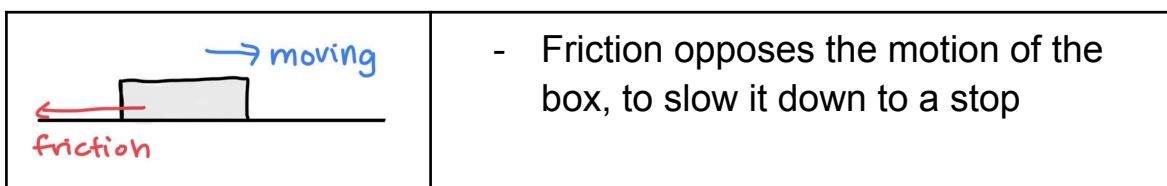
- The centripetal force is provided by the tension in the spring, and the weight of the ball
- So, when resultant force needs to increase to keep the ball moving in a circular path, the tension of spring increases
- The spring breaks when the tension increases until it can no longer support the resultant force.
- The ball will escape from the circular path along a tangent line at the point of breakage, then it is pulled downwards to the ground because of weight



Friction

Solid friction: the force between two surfaces that may impede motion and produce heating

- Friction is the force that opposes the movement of an object, and acts in the direction opposite to the motion
- Friction slows down objects, as it converts the kinetic energy into thermal energy, which is lost to the surroundings



Fluid friction: (drag force)

- Friction can also act on an object moving through a fluid -> this is air or liquid resistance
- When surface area increases, drag increases
- When speed increases, drag increases

Streamline shapes:

- Streamline shapes experience less drag, because they deflect the fluid more smoothly, resulting in less resistance to the object's movement

1.5 Turning effects of forces

Moment

- Turning effects of forces = moment of forces
- Examples of moment: seesaw, wrench, lever

Moment of a force: force x perpendicular distance from pivot

Moment of a force about a pivot: a measure of its turning effect

$$M = Fd$$

moment (Nm) force (N) perpendicular distance from pivot (m)

Example - calculate total moment about P

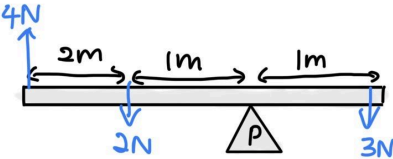


Diagram showing a horizontal beam with a pivot P. A 4N force acts upwards at the left end, 2m from the pivot. A 2N force acts downwards at the left end, 1m from the pivot. A 3N force acts downwards at the right end, 1m from the pivot.

$$M_c = (3 \times 1) + (4 \times 3)$$
$$= 3 + 12 = 15 \text{ Nm}$$
$$M_{Ac} = 2 \times 1 = 2 \text{ Nm}$$
$$M_{\text{total}} = 15 - 2$$
$$= 13 \text{ Nm clockwise}$$

Principal of moment and Equilibrium

Principal of moment - When object is in equilibrium, the total clockwise moment is equal to the total anticlockwise moment

Conditions for equilibrium:

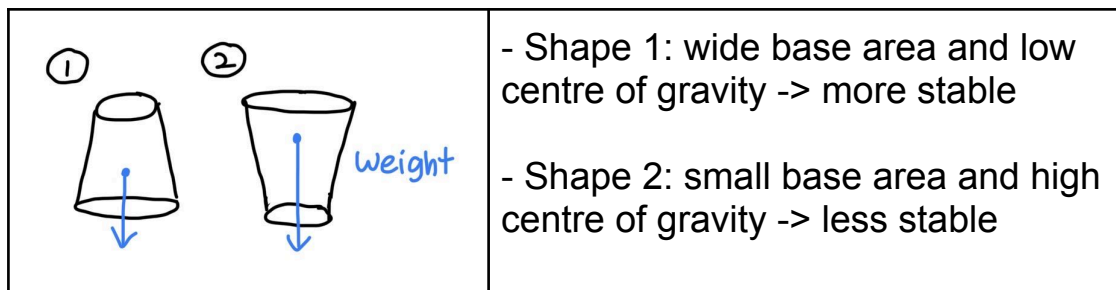
- No resultant force acting on object
- No resultant moment acting on object

Note -> total upwards force = total downwards force

Centre of gravity

Centre of gravity: the point where all weight acts

- An object will always balance around its centre of gravity
- An object is stable if it has a low centre of gravity and a wide base area



- An object will topple over if a vertical line from its centre of gravity falls outside of its base area



Experiment to investigate the centre of gravity of irregular piece of paper / lamina:

- Suspend a piece of paper from a pin through a hole near its edge
- Hang a plumb line (thread with weight) from the same pin, so it hangs vertically in front of the paper
- Use a ruler to draw a straight line on the paper, that follows the plumb line
- Repeat process at 2 other holes at 2 other positions
- The point where all 3 lines intersect is the centre of gravity

1.6 Momentum

Momentum - mass x velocity

- It is the measure of the tendency of an object to keep moving (how hard it is to stop moving)

$$p = mv$$

momentum (kgm/s or Ns) mass (kg) velocity (m/s)

Impulse - force x time for which force acts

- Also known as change in momentum

$$\text{Impulse} = Ft = m(v-u)$$

- Force $\times$ time
- change in momentum

Resultant force - the change in momentum per unit time

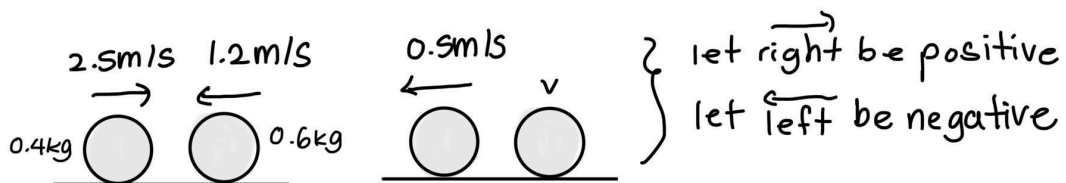
$$F = \frac{m(v-u)}{t} \quad \begin{array}{l} \text{impulse (kgms or Ns)} \\ \text{time (s)} \end{array}$$

force (N)

Momentum in collision:

- Conservation of momentum - total momentum before collision is equal to total momentum after collision

Example - find speed of B after collision



Before collision:

$$\begin{aligned} p &= mv \\ &= (0.4 \times 2.5) + (0.6 \times -1.2) \\ &= 0.28 \text{ kgms} \end{aligned}$$

After collision:

$$\begin{aligned} p &= (0.4 \times -0.5) + 0.6v \\ &= -0.2 + 0.6v \end{aligned}$$

momentum(before) = momentum(after)

$$0.28 = -0.2 + 0.6v$$

$$0.6v = 0.48$$

$$v = \frac{0.48}{0.6} = 0.8 \text{ m/s}$$

So, $v = 0.8 \text{ m/s}$ to the right

Note - be careful when doing problems involving vector quantities -> choose right to be positive and left to be negative

Momentum in safety car:

- Larger force means faster change of momentum
 - In a car crash, momentum changes very quickly, so forces on the body will be very large
-

1.7 Energy, work and power

Energy

- the ability of a body to do work
- Scalar quantity
- Unit: Joule (J) or Newton metre (Nm)

Kinetic energy: energy possessed by a mass, due to its movement

$$E_k = \frac{1}{2}mv^2$$

kinetic energy (J) mass (kg) velocity (m/s)

Gravitational Potential energy: energy stored in a mass, due to its position in a gravitational field

$$E_p = mgh$$

gravitational potential energy (J) mass (kg) height (m) gravitational field strength (N/kg)

Energy may be stored as:

- kinetic
- gravitational potential
- chemical
- Elastic (strain)
- Nuclear

- Electrostatic
 - Internal (thermal)
-

Work

- Scalar quantity
- Unit: Joules (J) or Newton metre (Nm)

$$W = Fd$$

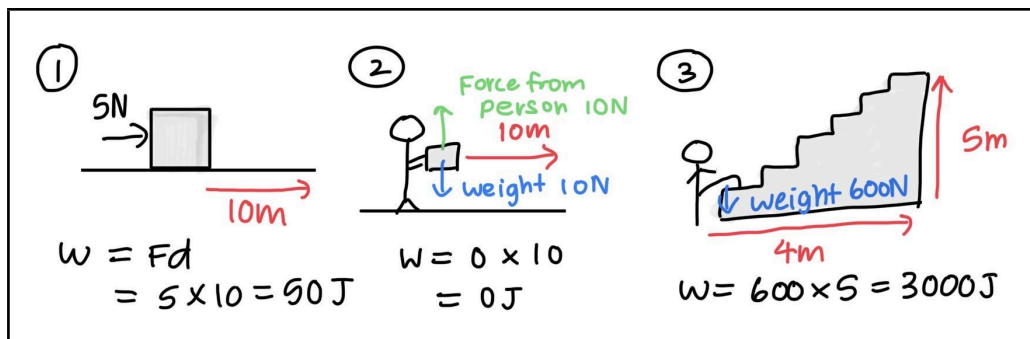
distance (m)
force (N)

work (J)

- Product of force and distance moved in the direction of the force

Examples:

- always note has to be 'distance moved in the direction of the force'



Work done - Energy principle

- Work done is equal to energy transferred ($W = Fd = \Delta E$)
- So, they may ask you for energy, and it's using this formula.

Example 1 - asked to find velocity of box

Position 1 Position 2

* the surface does not cause thermal energy loss

$$W = Fd = 5 \times 10 = 50 \text{ J}$$

$$E(P1) = \frac{1}{2}mv^2 = \frac{1}{2}(5)(2)^2 = 10 \text{ J}$$

$$E(P2) = \frac{1}{2}(5)v^2 = 2.5v^2$$

$$\Delta E = 2.5v^2 - 10$$

$$2.5v^2 - 10 = 50$$

$$2.5v^2 = 60$$

$$v^2 = 24 \quad v = \sqrt{24} = 4.9 \text{ m/s} \checkmark$$

- Note: the ΔE has to be energy (P2) - energy (P1)
- This is because when a resultant force is applied, object accelerates in direction of resultant force
- So, the velocity of P2 will be higher, meaning its energy is also higher

Example 2 - asked to find the distance travelled

Position 1 Position 2

Surface causes thermal energy loss = 5J

$$W = Fd = 10d$$

$$E(P1) = \frac{1}{2}mv^2 = \frac{1}{2}(5)(2)^2 = 10 \text{ J}$$

$$E(P2) = \frac{1}{2}(5)(5)^2 = 62.5 \text{ J}$$

$$\Delta E = 62.5 - 10 = 52.5$$

$$\text{actual energy transfer} = 52.5 + 5 = 57.5 \text{ J}$$

$$10d = 57.5$$

$$d = 5.75 \text{ m} \checkmark$$

- You must add the energy loss, as that is also part of the energy transferred

Principal of conservation of energy

- Energy cannot be created or destroyed, and can only be transferred from one form to another

Situation 1: when a ball is dropped from rest

- $KE = 0$ because speed = 0m/s
- GPE is maximised, greatest height
- As it is dropped, its height decreases
- GPE decreases because height decreases
- GPE is transferred to KE, which causes speed to increase
- Also transferred to heat, which is lost to the surroundings
- When it reaches the ground, height = 0, so $GPE = 0$
- All GPE is transferred to KE and heat

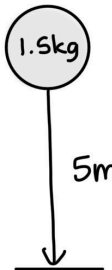
Situation 2: when a ball is thrown into the air

- $GPE = 0$, because height = 0
- KE is maximised, greatest speed
- As it moves upwards, speed decreases
- KE decreases because speed decreases
- KE is transferred to GPE, which is why height increases
- KE is also transferred to heat, which is lost to the surroundings
- When it reaches the highest point, speed = 0m/s , so $KE = 0$
- all KE is transferred to GPE and heat

- Energy transferred to heat is due to air resistance

air resistance is ignored

To find speed that it reaches the ground



1.5kg
5m

$$KE = GPE$$

$$\frac{1}{2}mv^2 = mgh$$

$$\frac{1}{2}v^2 = (9.8)(5)$$

$$= 49$$

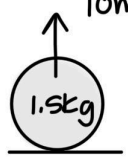
$$v^2 = 49 \div \frac{1}{2}$$

$$= 98$$

$$v = \sqrt{98}$$

$$= 9.9 \text{ m/s} \# \checkmark$$

To find maximum height — assuming 15J is lost to surroundings



10m/s
1.5kg

$$KE = \frac{1}{2}mv^2$$

$$= \frac{1}{2}(1.5)(10)^2$$

$$= 75\text{J}$$

$$GPE = KE - 15$$

$$= 60\text{J}$$

$$mgh = 60$$

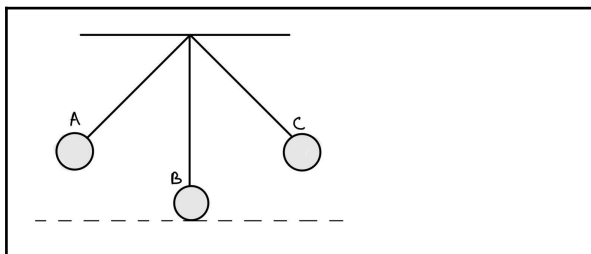
$$(1.5)(9.8)h = 60$$

$$14.7h = 60$$

$$h = 4.1\text{m} \# \checkmark$$

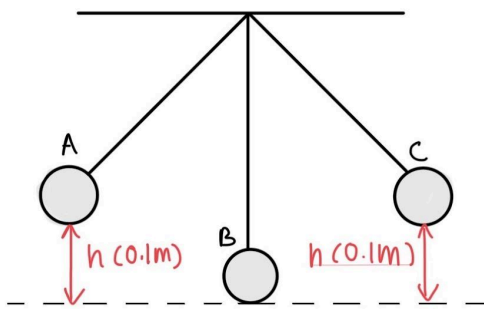
Other examples

Pendulum:



- At Point A, $KE = 0$ because speed = 0
- GPE is maximised, greatest height

- From Point A to Point B, height decreases, so GPE decreases
- GPE is transferred to KE, speed increases
- At Point B, GPE is minimised, lowest height
- KE is maximised, greatest speed
- From Point B to Point C, height increases, so GPE increases
- KE is transferred to GPE, speed
- At Point C, all KE is transferred to GPE.
- Speed = 0



maximum speed mass reaches at point B

- GPE does not decrease to 0 as height does not decrease to 0

using height difference: (from A to B)

$$\Delta \text{GPE} = \text{KE}$$

$$mg\Delta h = \frac{1}{2}mv^2 \quad v^2 = 1.96$$

$$(9.8)(0.1) = \frac{1}{2}v^2 \quad v = 1.4\text{m/s} \# \checkmark$$

A sloped surface:

A diagram showing a mass sliding down a rough surface. The mass starts at a height of 2m and ends at a height of 0.5m. The surface is rough, causing energy loss. The mass is labeled 1.5kg and is at rest at the start. The final speed is labeled v .

highest GPE
0 KE
GPE $\rightarrow$ KE
GPE $\rightarrow$ heat loss

rough surface causes energy loss
Find final speed

$$\Delta \text{GPE} = \text{KE} + 5$$

$$mg\Delta h = \frac{1}{2}mv^2 + 5$$

$$(1.5)(9.8)(1.5) = \frac{1}{2}(1.5)v^2 + 5$$

$$v = 4.8\text{m/s}$$

Power & Efficiency

Power: work done per unit time

- Scalar quantity
- Unit: Watts (W) or Joules per second (J/s)

$$\begin{array}{c} \text{power} \\ (W \text{ or } J/s) \end{array} P = \frac{W}{t} \begin{array}{c} \text{work (J)} \\ \text{time (s)} \end{array}$$

Efficiency: percentage of total energy input which is transferred into useful energy output

- Note -> efficiency can be expressed as a decimal, fraction or percentage

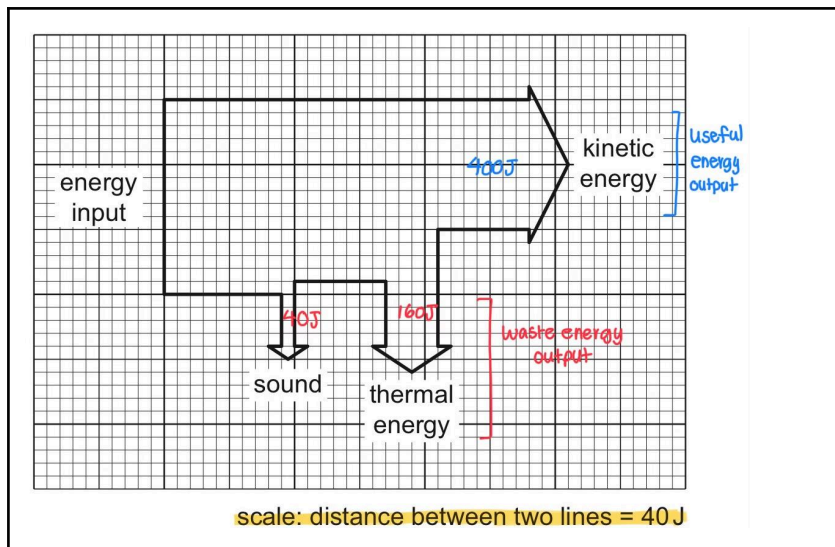
$$\text{Efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100$$

$$\text{Efficiency} = \frac{\text{useful power output}}{\text{total power input}} \times 100$$

Examples	Input energy	Output energy & Efficiency
Light bulb	Electrical energy = 120J	Useful = light = 50J Waste = heat = 70J Efficiency = $(50/120) \times 100 = 41.7\%$
Television	Electrical energy = 550J	Useful = light = 220J = sound = 250J Waste = heat = 80J Efficiency = $(470/550) \times 100 = 85.5\%$
Running person	Chemical potential energy = 800J	Useful = kinetic = 500J Waste = heat = 300J Efficiency = $(500/800) \times 100 = 62.5\%$

Sankey diagrams

- Used to represent energy transfers
- Remember, conservation of energy: total energy input = total energy output



- Note: usually don't have arrow heads (so don't include them when drawing sankey diagram)

Energy resources

Renewable energy: energy which can be replenished

Biofuels/Biomass -> waste of living things:

- Contains chemical potential energy
- Derived from the sun
- An example is wood (fuelwood)

Geothermal energy:

- thermal energy found beneath the earth's surface

Wind energy:

- Contains kinetic energy
- Derived from the sun

Hydroelectric energy:

- Contains gravitational potential energy
- Derived from the sun

Tidal energy:

- Contains gravitational potential energy
- Derived from the moon

Wave energy:

- Contains kinetic energy
- Derived from the sun

Solar energy:

- Light energy from the sun
-

Non-renewable energy: energy which cannot be replenished

Fossil fuel -> coal, oil and natural gas:

- Contains chemical potential energy
- Derived from the sun

Nuclear fuel -> uranium and plutonium:

- Contains nuclear energy

So, in conclusion,

Derived from the sun:

- | | |
|------------------------|----------------|
| - Biofuels/biomass | - Wave energy |
| - Wind energy | - Solar energy |
| - Hydroelectric energy | - Fossil fuels |

Derived from the moon -> tidal energy

Not derived from the sun or moon:

- Geothermal energy
- Nuclear energy

1.8 Pressure

Pressure on a solid

Pressure: force per unit area

$$p = \frac{F}{A}$$

pressure (Pa) force (N) area (m²)

Pressure increases as:

- Contact area increases (while force remains constant)
- Force increases (while contact area remains constant)

Applications of pressure in everyday life:

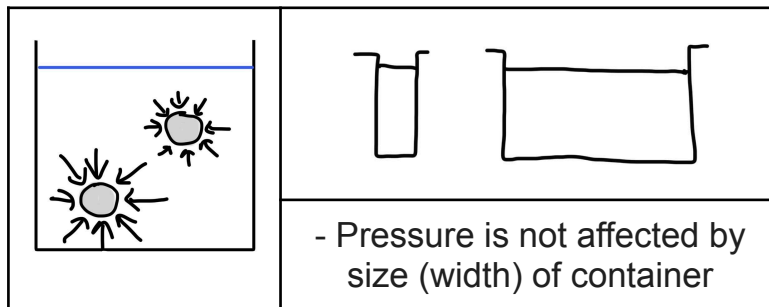
- Spikes - small area to increase pressure, to grip the ground better
- Scissors, sharp tools - small area to increase pressure, to cut object better
- Shovel - small area to increase pressure, to dig the soil better
- Big feet for animals - large area to reduce pressure, so that they do not sink deeply into soft ground, such as snow

Pressure in a liquid

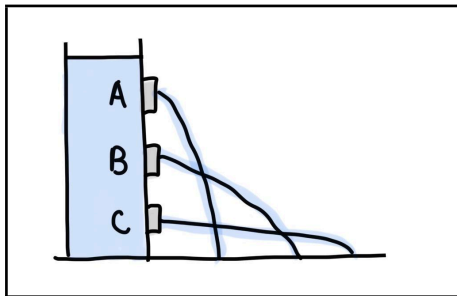
- Pressure in a liquid acts in an object in all directions
- Pressure is higher at greater depths

$$p = \rho g h$$

pressure (Pa) density (kg/m³) gravitational field strength (N/kg) height (m)



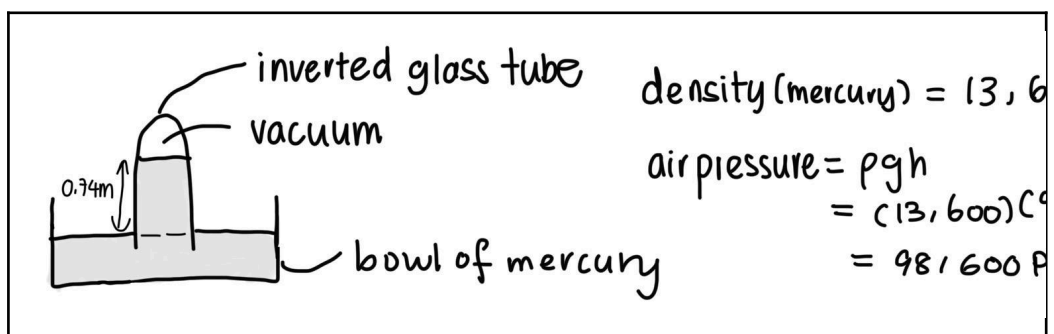
Container with 3 holes, and liquid coming out of it:



- Liquid flows out of A over a shorter distance, because point A is shallower, lowest depth, and has the lowest pressure

Barometer

- Barometer is used to measure air pressure



- Air pressure = mercury pressure in column
 - *If not vacuum, air particles at the top of the column can reduce mercury column height, which causes barometer to underestimate air pressure
-